01 DATA

02 INPUTS

03 MODELS

04 PREDICTION

CGMacros
official

45 adults · 1,397 meals · ~10 days





Meal photos · CGM · Heart rate · Activity · Food log · Age · sex · BMI

days 1–7 · train · 8+ · test

MetaboNet
official

1,085 people with type 1 diabetes · 671k meals

CGM · Carbs · Insulin · Age · sex · BMI

BIG IDEAs

16 adults · 397 meals

CGM · Food log · Sex

Meal photo
CGMacros



Image encoder
MobileCLIP-S2 · frozen



Nutrition estimator
nearest meals + GBM



512-d

carbs · protein · fat · fiber · kcal

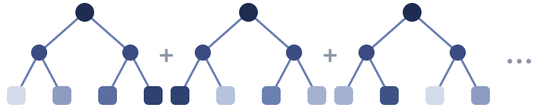
- Nutrients from photo
- Glucose before the meal
- Meal time · previous meal
- First-week glucose profile
- Similar meals, own and others'
- Past meals with their glucose curves
- Heart rate
- Activity (METs, active kcal)
- Age · sex · BMI
- Logged kcal (optional)
- Insulin dose and timing
- Glucose after the meal (target)

A B

MODEL A

Gradient-boosted trees

LightGBM · glucose at 15–120 min, peak, iAUC



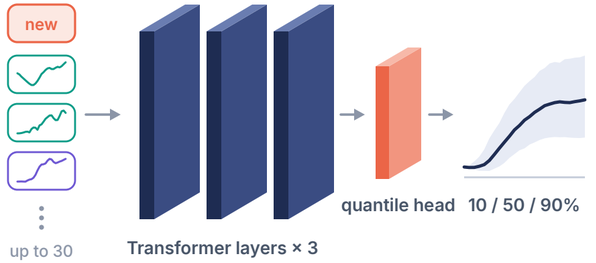
Trained on · CGMacros · days 1–7 · BIG IDEAs

MODEL B

In-context transformer

reads the person's past meals, outputs the whole curve

meal tokens



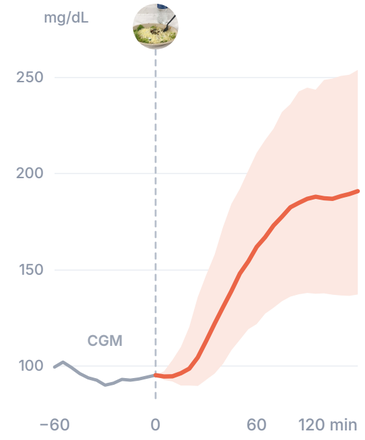
1 Pretrain · MetaboNet, 671k meals

2 Fine-tune · CGMacros, days 1–7

Σ blend

Glucose response

next 2 hours · 80% band



Peak glucose	Peak rise
191 mg/dL	+97 mg/dL
real 187	real +92

Time to peak	iAUC, mg/dL·min
90 min	6.2k
real 106	real 6.3k

Example: a real lunch on an unseen test day; "real" is the recorded CGM.

PROFILE MODELS

Colas 2019

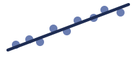
208 adults · 2.7-year follow-up

CGM, 1–2 days · Clinical labs · T2D outcome

- CGM metrics (up to 7 days)
- Age · BMI · fasting glucose · HbA1c
- Lab HbA1c (target)
- Developed type 2 diabetes (target)

HbA1c estimate

Ridge regression · 261 people



→ Estimated HbA1c

Type 2 diabetes risk

Logistic regression · 208 people



→ Risk tier